ARKOPROBHO MITRA

Senior AI Engineer | Generative AI, RAG & Agentic Systems

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PROFESSIONAL SUMMARY

AI Engineer with nearly 3 years developing GenAI and applied ML solutions within 5+ years at Infosys. Contributed to enterprise RAG, document intelligence, fraud detection, and multi-agent automation for regulated domains. Experienced in developing secure, evaluated AI systems, integrating enterprise tools, and improving retrieval quality, workflow efficiency, and inference costs.

TECHNICAL SKILLS

GenAI and agents: Azure OpenAI (GPT-3.5 Turbo, GPT-4, GPT-4o), LangGraph, LangChain, MCP, RAG, hybrid retrieval, RRF, Azure AI Search
Evaluation and security: RAGAS, SME golden sets, Langfuse, CI/CD gates, Azure AD, Key Vault, human review, audit logging, MLflow, Evidently AI
ML and engineering: Python, scikit-learn, XGBoost, SHAP, LayoutLMv2, OCR, Azure Document Intelligence, FastAPI, asyncio, PostgreSQL, Redis, Docker, GitHub Actions, Elasticsearch, SAP Ariba, ServiceNow, SharePoint

PROFESSIONAL EXPERIENCE

Infosys | Aug 2021 - Present

- Design and develop a smart contractual interface using **Generative AI** and **Model Context Protocol (MCP)** to support contract-related workflows.
  - Integrate **MCP** tools into the interface to support dashboard functionality and workflow execution.
  - Evaluate **MCP** tool behavior against functional requirements to identify gaps and improvement opportunities.
  - Develop and refine **MCP** tools to address identified gaps and meet contractual interface requirements.
  - Prepare tool-performance reports and improvement recommendations to guide ongoing development.
- Developed a **five-agent** procurement copilot using **LangGraph** and **Azure OpenAI GPT-4o**; integrated SAP Ariba, ServiceNow, SharePoint, and contract **RAG**, reducing procurement cycle time by 35% and RFQ turnaround from three days to four hours.
- Contributed to an insurance claims platform using **LayoutLMv2** for **document understanding** and **XGBoost** with **SHAP** for explainable **fraud detection**; achieved 94% **document classification** accuracy and reduced claim resolution from four days to about two and a half.
- Developed **hybrid retrieval**, **access controls**, and **evaluation** for an enterprise **RAG** assistant over 2.5 million documents; reduced compliance-query time by about 60% in a pilot and supported scaling to 30,000+ monthly users.
- Developed and optimized logistics **data pipelines** and **Elasticsearch** search; built **QlikSense** dashboards and contributed to ServiceNow incident workflows, client **data migrations**, and legacy-system integrations.

RESEARCH PUBLICATION

How to handle bi/tri-lingual Indic texts in a single image? A new dataset of natural scene and born-digital images
Multimedia Tools and Applications, Springer, Vol. 81, pp. 15367-15394 (2022)
Co-authored a multilingual image dataset for Indic OCR and scene-text detection. DOI: 10.1007/s11042-022-12596-7

SELECTED PORTFOLIO PROJECTS

Agentic Job Application Pipeline | 2026: Built a LangGraph, FastAPI and asyncio workflow for discovery, LLM scoring, application drafting and human review.
LangGraph Fact-Checker | 2026: Built a FAISS verification pipeline with confidence-based human review and Langfuse observability.
E-Commerce Platform with RAG Assistant | 2025: Built React, Spring Boot and FastAPI services with semantic search, sentiment analysis and RAG support.

EDUCATION AND CERTIFICATION

Bachelor of Engineering | Jadavpur University, Kolkata | 2017 - 2021
AWS Certified Cloud Practitioner | Amazon Web Services